

This writeup aims to explain I-splines, which are a form of monotone B-splines.

1 M-Splines

Fix a knot sequence $\mathbf{t} = [t_1, \dots, t_{n+k}]$ where n is the number of free parameters and k is the polynomial order of the spline. Let us use the recursive definition of M-splines, given by the following.

Definition 1.1 (M-splines). *An M-spline $M_{i,k,t}(x)$ is given by the following recursion.*

1. $M_{i,1,t}(x) = \begin{cases} \frac{1}{t_{i+1}-t_i}, & t_i \leq x < t_{i+1} \\ 0, & \text{otherwise} \end{cases}$
2. $M_{i,k,t}(x) = \frac{k[(x-t_i)M_{i,k-1,t}(x) + (t_{i+k}-x)M_{i+1,k-1,t}(x)]}{(k-1)(t_{i+k}-t_i)}$

Note that at knots where $t_{i+k} = t_i$, the term $\frac{0}{0}$ is taken as 0 by convention.

One can note the following properties of M-splines which follows from the definition.

1. $\int M_i(x) = 1$,
2. $M_{i,k,t}(x)$ is strictly positive only when $t_i \leq x < t_{i+k}$.
3. M-splines are non-negative everywhere.

Proposition 1. *A B-spline is related to an M-spline by the following relationship,*

$$B_{i,k,t} = \frac{(t_{i+k} - t_i)M_{i,k,t}}{k}$$

Proof. Follows by the recursive definitions of B-splines and M-splines. ■

2 I-splines

Definition 2.1 (I-spline). *An I-spline is defined as*

$$I_{i,k,t}(x) = \int_L^x M_{i,k,t}(u) du$$

where L is the left end of the support.

From this definition, we can also write it in the recursive form, as follows, For $t_j \leq x < t_{j+1}$,

$$I_{i,k,t}(x) = \begin{cases} 0, & i > j, \\ \sum_{m=i}^j \frac{t_{m+k+1}-t_m}{k+1} M_{m,k+1,t}(x), & j - k + 1 \leq i \leq j, \\ 1, & i < j - k + 1. \end{cases}$$

It is easy to see the most important property of I-splines, that is it is increasing in x , which is obvious by construction since M-splines are non-negative.

Using Proposition 1, it is also easy to write I-splines in terms of B-splines.

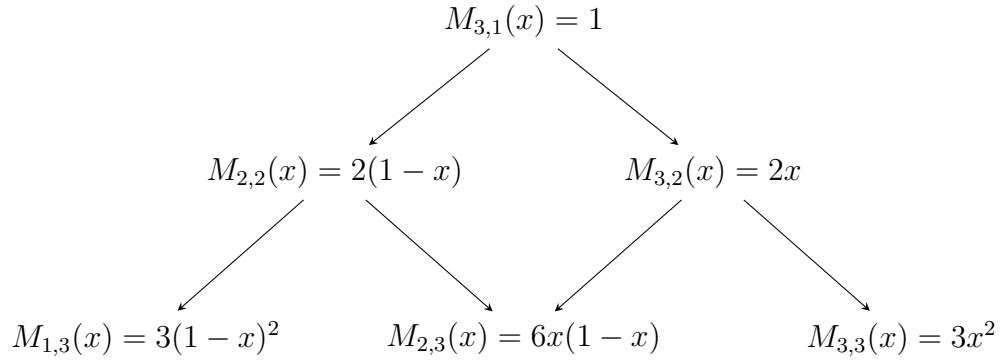
3 Simple Example

Fix $L = 0, U = 1$, order $k = 3$ (piecewise quadratic M-splines, cubic I-splines), and no interior knots. The knot vector is:

$$t = \{t_1, t_2, t_3, t_4, t_5, t_6\} = \{0, 0, 0, 1, 1, 1\}, \quad n = 3.$$

The only non-empty active interval is $x \in [t_3, t_4) = [0, 1)$.

Next, recursively build the M-splines:



Finally, we can construct the I-Splines by direct integration.

Using $I_{i,3,t}(x) = \int_0^x M_{i,3,t}(u) du$ on $[0, 1]$:

$$\begin{aligned} I_1(x) &= \int_0^x 3(1-u)^2 du = \left[-(1-u)^3 \right]_0^x = 1 - (1-x)^3, \\ I_2(x) &= \int_0^x 6(u-u^2) du = \left[3u^2 - 2u^3 \right]_0^x = 3x^2 - 2x^3, \\ I_3(x) &= \int_0^x 3u^2 du = \left[u^3 \right]_0^x = x^3. \end{aligned}$$

Outside $[0, 1]$, each $I_i(x)$ is constant: 0 for $x \leq 0$ and 1 for $x \geq 1$.

Now let us use the above I-splines for a simple regression problem. Consider a DGP given by $x_i \sim \text{Uniform}[0,1]$ and $y_i = 0.5 + 2x^2 + \epsilon_i$ where $\epsilon_i \sim N(0,0.1)$. Using the above I-spline basis, we run the following regression specification

$$y_i = a_0 + \sum_{m=1}^3 a_m I_m(x_i) + \varepsilon_i, \quad \text{subject to } a_m \geq 0 \text{ for } m \in \{1, 2, 3\}$$

and optimize by a constrained least squares problem given by

$$\begin{aligned} \min_{\mathbf{a} \in \mathbb{R}^4} \quad & \|\mathbf{y} - \Phi \mathbf{a}\|_2^2 \\ \text{s.t.} \quad & a_0 \in \mathbb{R}, \\ & a_1, a_2, a_3 \geq 0 \end{aligned}$$

where

$$\Phi = \begin{bmatrix} 1 & I_1(x_1) & I_2(x_1) & I_3(x_1) \\ 1 & I_1(x_2) & I_2(x_2) & I_3(x_2) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & I_1(x_N) & I_2(x_N) & I_3(x_N) \end{bmatrix}.$$

Here is the resulting estimate in a plot.

